Full Stack Web Developer - Technical Assessment

Motorola Solutions Web Development Team

Overview

Your objective is to build a full-stack Next.js catalog application using a **Multi-Agent Development System**.

Rather than testing your ability to manually type React boilerplate, we are evaluating your ability to architect LLM pipelines, manage complex agentic state, and enforce strict coding requirements through **Markdown-driven agent instructions**. You will define a "team" of specialized AI agents via Markdown files and use them to generate the codebase.

Part 1: The Target Output (The App Spec)

This is the specification your agent system must be prompted to build.

Target Stack: Next.js, Contentful, TypeScript, Tailwind CSS

Core Application Requirements:

- **Simulated Authentication:** A full-stack login flow. No real database validation needed (accept any non-empty input). The app must securely set a session/cookie on the server and use it to protect routes.
- **Role-Based Content & Pricing:** Two product card sets (Set A: public, Set B: authenticated only). Each product has `priceLogged0ut` and `priceAuthenticated`.
- **Rendering:** Pricing and role-based content must be server-side rendered (SSR/SSG) in the initial HTML, not client-only.
- **Display Mapping:** Support 2+ layout options (e.g., “Price First”, “Specs First”). The user can switch layouts via a UI control without page reload.
- **Filtering:** Filter by category, text search, and a “Clear Filters” button.
- **Error Handling:** Graceful fallbacks for missing product images, empty search results, and invalid slugs.

Part 2: The Core Task (The Markdown Agent System)

This is the methodology you must use to build the app.

Design a multi-agent system using **Markdown instruction files** (e.g., system prompts, persona definitions, or chained instructions) that direct an AI assistant (such as GitHub Copilot, Cursor, or ChatGPT) to build the functional application specified in Part 1.

Requirements & Constraints:

- **Prompt-Driven Architecture:** Design your AI team entirely through Markdown files. You have total architectural autonomy over how many agents you create (e.g., a "Frontend Architecture Agent", a "State Management Agent", etc.), how they hand off context to each other, and how they divide the labor.
- **Agentic Best Practices:** Do not use a single monolithic "write the whole app" prompt. Break the problem down into specialized roles and discrete steps via your Markdown instructions.
- **Output Structure:** Organize your agent definitions cleanly in a dedicated directory (e.g., `/agents/` or `/prompts/` in your repository) so we can review your prompting strategy and system boundaries.

> **Note:** While the app needs to integrate with Contentful, your agents can be instructed to mock the Contentful data fetch layer with a hardcoded JSON seed file to keep the generation scope realistic.

Deliverables

- **Repository:** A single repository containing your Next.js application codebase AND your `/agents/` directory with the Markdown prompt files.
- **README.md:**
  - Setup instructions to run the Next.js app locally.
  - Deployment URL of the generated app (Vercel/Netlify).
- **AGENT.md:**
  - **Architecture Explanation:** Walk us through your multi-agent architecture. Why did you define the specific agents/prompts that you did? How did you manage context and handoffs between the different prompts?
  - **Prompting Strategy:** How did you enforce the specific architectural requirements (like SSR vs Client-side rendering) through your markdown instructions?
  - **Trade-offs & Manual Intervention:** What manual fixes (if any) were required after the AI ran to get the app to compile and deploy? Discuss where the AI struggled and what you'd improve if given two more weeks.

Submission

To: jeffrey.masterson@motorolasolutions.com
Subject: Full Stack Developer Assessment - [Your Name]

Include in the email body:

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Repo: [url]
Live App: [url]
Demo Login: [credentials if needed]
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Questions? Email us within 24 hours of receiving the assignment.

Good luck!
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